Depropanizer Column Ver 1 · OTS

DC3-V1-CM-PCC-001

Pre-commissioning checklist

Rev 0 · Issued for teaching

Unit	Depropanizer Column Ver 1 · OTS
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Revision record

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Note. *Numbers come from the simulator model (design case) and the assumptions listed; not for construction. Codes are named for the method only; no client, contractor, licensor or vendor is implied.*

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1 · Purpose, systems and completion categories

Pre-commissioning checklist of the C-201 unit, generated from the equipment list (design pressures), the line list, the instrument list, the load list, the relief basis and the C&E matrix. Each row is signed by construction, commissioning and the owner; when every row of a system is signed, the system is ready for commissioning. Mechanical completion is certified system by system after a walk-down; punch items are classified A (pressure integrity, safety or the ability to test: before the MC certificate), B (before hydrocarbon or heat is introduced) and C (cosmetic or documentation: before final acceptance). In this document the hydrotest and flushing are pre-commissioning activities; many contracts place them before MC.

System	Name	Scope	Items
S10	Feed system	V-201, P-201A/B, E-201 tube side and the feed lines	12
S20	Column and reboiler	C-201, E-203 tube side, sump and thermosyphon lines	9
S30	Overhead and reflux	E-202 shell side, V-202, P-202A/B, reflux, product and vent lines	17
S40	LP steam and condensate	E-201 and E-203 shell sides, steam and condensate lines	10
S50	Cooling water	E-202 tube side and the CW lines	5
S60	Relief to flare	PSV-201 and its isolation valves	1
S70	Instruments, DCS and SIS	field instruments, analyzers, DCS, SIS	58
S80	Electrical	motors, MCC, UPS, lighting	3

Category	Items
FL · Flushing and blowing	20
HT · Hydrotest	21
DR · Drying	3
PG · N_2 purge (O_2 < 2 %)	3
LT · Tightness test	5
VS · Valve stroke and fail position	10
LC · Instrument loop check	42
PS · PSV bench test and set certificate	1
MB · Motor bump test / turbine solo run	4
SF · SIS function test	6
Total	115

2 · Hydrotest pressures

Worked calculation, C-201

- UG-99(b): P_T = 1.3 × MAWP × stress ratio; MAWP = design, stress ratio 1 (assumed). Top section 1.90 → 2.470 MPa g; bottom section 1.95 → 2.535 MPa g.
- Shop test horizontal: top gauge 2.535 MPa g so every course sees its own test pressure.
- Field test vertical: top gauge 2.470 MPa g + water head 1000 × 9.81 × 23.82 m = 234 kPa → bottom 2.704 MPa g ≥ 2.535 ✓.
- Bottom hoop stress σ = P (R + 0.6 t) / t = 2.704 × (850 + 10.6) / 17.7 = 131.5 MPa ≤ 0.9 S_y = 234 MPa ✓.
- Water weight: 53.4 t of water + 23.2 t steel = 76.6 t, against 27.6 t in operation (query EC-03).

Item	Side	System	Design MPa g	Hydrotest MPa g	Water m³	Note
C-201	Shell — top	S20	1.90	**2.470**	53.42	
C-201	Shell — bottom	S20	1.95	**2.535**	53.42	
V-201	Shell	S10	1.35	**1.755**	24.94	
V-202	Shell	S30	1.90	**2.470**	15.78	
E-201	Shell (LP steam)	S40	0.70	**0.910**	—	each side separately
E-201	Tubes (feed)	S10	2.85	**3.705**	—	each side separately
E-202	Shell (overhead)	S30	1.90	**2.470**	—	each side separately
E-202	Tubes (cooling water)	S50	0.70	**0.910**	—	each side separately
E-203	Shell (LP steam)	S40	0.70	**0.910**	—	each side separately
E-203	Tubes (C-201 bottoms)	S20	1.95	**2.535**	—	each side separately
P-201A/B	Casing	S10	2.85	**3.705**	—	vendor casing test API 610 4.275 MPa g
P-202A/B	Casing	S30	2.60	**3.380**	—	vendor casing test API 610 3.900 MPa g

Test pack	System	Class	Lines	Design MPa g	Hydrotest MPa g (1.5 ×)
TP-10-01	S10 Feed system	A1L	6"-P-200, 6"-P-201	1.35	**2.025**
TP-10-02	S10 Feed system	B1L	4"-P-202, 6"-P-203	2.85	**4.275**
TP-20-01	S20 Column and reboiler	B1L	8"-P-211, 10"-P-212	1.95	**2.925**
TP-30-01	S30 Overhead and reflux	B1L	8"-P-204, 6"-P-205, 1"-P-206, 6"-P-207	1.90	**2.850**
TP-30-02	S30 Overhead and reflux	B1L	4"-P-208, 4"-P-209, 3"-P-213	2.60	**3.900**
TP-30-03	S30 Overhead and reflux	B1L	4"-P-210	1.95	**2.925**
TP-40-01	S40 LP steam and condensate	A1S	6"-LS-301, 6"-LS-302	0.70	**1.050**
TP-40-02	S40 LP steam and condensate	A1C	2"-CD-501, 2"-CD-502	0.70	**1.050**
TP-50-01	S50 Cooling water	A1W	10"-CW-401, 10"-CW-402	0.70	**1.050**

3 · Drying, N_2 purge and tightness test

Pressure-swing purge with N_2 between 500 and 20 kPa g; each cycle multiplies the oxygen by (P_low + P_atm) / (P_high + P_atm). The page calculator varies the pressures and the target.

System	Volume m³ (+15 %)	Ratio per cycle	Cycles	O_2 after	N_2 Nm³	Continuous sweep Nm³
S10 Feed system	28.7	0.2017	2	0.95 %	245	62

System	Volume m³ (+15 %)	Ratio per cycle	Cycles	O_2 after	N_2 Nm³	Continuous sweep Nm³
S20 Column and reboiler	61.4	0.2017	2	0.95 %	524	133
S30 Overhead and reflux	18.1	0.2017	2	0.95 %	155	39

System	Max operating MPa g	90 % PSV set	Lowest design	Test pressure MPa g	Limited by
S10 Feed system	2.563	— (no PSV-201)	1.35	**0.80**	N_2 header
S20 Column and reboiler	1.740	1.62	1.90	**0.80**	N_2 header
S30 Overhead and reflux	2.355	1.62	1.90	**0.80**	N_2 header

4 · Pump run-in on water

Same head in metres, so pressure and power scale with density (pump curves of the model).

Pump	ρ kg/m³	× on water	kW at normal flow LPG → water	Motor kW	Max water flow m³/h	Shut-off MPa g LPG → water	Design MPa g	Result
P-201A	548.5	1.82	49.7 → 90.7	75	46	2.723 → 4.22	2.85	✗ overpressure motor overload
P-202A	471.7	2.12	20.7 → 43.9	30	47	2.421 → 1.51	2.60	✗ motor overload

No water run-in against the plant system (queries EC-01, EC-02).

5 · Checklist

FL · Flushing and blowing (20 items)

Item	Sys	Equipment / line / loop	Activity	Acceptance	Ref.	Done	Date / sign
FL-001	S10	6"-P-200	Water flushing (instruments and control valves removed or spooled)	Clean water at the outlet; temporary strainers clean	DC3-V1-PE-LIN-001	☐	
FL-002	S10	6"-P-201	Water flushing (instruments and control valves removed or spooled)	Clean water at the outlet; temporary strainers clean	DC3-V1-PE-LIN-001	☐	
FL-003	S10	4"-P-202	Water flushing (instruments and control valves removed or spooled)	Clean water at the outlet; temporary strainers clean	DC3-V1-PE-LIN-001	☐	
FL-004	S10	6"-P-203	Water flushing (instruments and control valves removed or spooled)	Clean water at the outlet; temporary strainers clean	DC3-V1-PE-LIN-001	☐	
FL-005	S30	8"-P-204	Air blowing from the vessel (quick-opening blow-down)	Clean white target (wooden board) after the blow	DC3-V1-PE-LIN-001	☐	
FL-006	S30	6"-P-205	Water flushing (instruments and control valves removed or spooled)	Clean water at the outlet; temporary strainers clean	DC3-V1-PE-LIN-001	☐	
FL-007	S30	1"-P-206	Air blowing from the vessel (quick-opening blow-down)	Clean white target (wooden board) after the blow	DC3-V1-PE-LIN-001	☐	
FL-008	S30	6"-P-207	Water flushing (instruments and control valves removed or spooled)	Clean water at the outlet; temporary strainers clean	DC3-V1-PE-LIN-001	☐	
FL-009	S30	4"-P-208	Water flushing (instruments and control valves removed or spooled)	Clean water at the outlet; temporary strainers clean	DC3-V1-PE-LIN-001	☐	
FL-010	S30	4"-P-209	Water flushing (instruments and control valves removed or spooled)	Clean water at the outlet; temporary strainers clean	DC3-V1-PE-LIN-001	☐	
FL-011	S30	4"-P-210	Water flushing (instruments and control valves removed or spooled)	Clean water at the outlet; temporary strainers clean	DC3-V1-PE-LIN-001	☐	
FL-012	S20	8"-P-211	Water flushing (instruments and control valves removed or spooled)	Clean water at the outlet; temporary strainers clean	DC3-V1-PE-LIN-001	☐	
FL-013	S20	10"-P-212	Water flushing (instruments and control valves removed or spooled)	Clean water at the outlet; temporary strainers clean	DC3-V1-PE-LIN-001	☐	
FL-014	S40	6"-LS-301	Steam blowing to the target plates (silencer at the free end)	Two consecutive polished target plates without pitting (assumed criterion)	DC3-V1-PE-LIN-001	☐	
FL-015	S40	6"-LS-302	Steam blowing to the target plates (silencer at the free end)	Two consecutive polished target plates without pitting (assumed criterion)	DC3-V1-PE-LIN-001	☐	
FL-016	S50	10"-CW-401	Water flushing at ≥ 1.5 × the design velocity, temporary strainer before E-202	Clean water at the outlet; strainer clean after 30 min	DC3-V1-PE-LIN-001	☐	
FL-017	S50	10"-CW-402	Water flushing at ≥ 1.5 × the design velocity, temporary strainer before E-202	Clean water at the outlet; strainer clean after 30 min	DC3-V1-PE-LIN-001	☐	
FL-018	S30	3"-P-213	Water flushing (instruments and control valves removed or spooled)	Clean water at the outlet; temporary strainers clean	DC3-V1-PE-LIN-001	☐	
FL-019	S40	2"-CD-501	Flush with LP steam / condensate to drain before the trap	Clean condensate at the drain; trap strainers cleaned	DC3-V1-PE-LIN-001	☐	
FL-020	S40	2"-CD-502	Flush with LP steam / condensate to drain before the trap	Clean condensate at the drain; trap strainers cleaned	DC3-V1-PE-LIN-001	☐	

HT · Hydrotest (21 items)

Item	Sys	Equipment / line / loop	Activity	Acceptance	Ref.	Done	Date / sign
HT-001	S20	C-201 shell — top	Hydrotest at 2.470 MPa g (top), hold 30 min (assumed), then drain fully	No leak, no visible deformation; gauge calibrated; test pressure ≥ 1.3 × design 1.90 MPa g	DC3-V1-ME-EQL-001	☐	
HT-002	S20	C-201 shell — bottom	Hydrotest at 2.535 MPa g (top), hold 30 min (assumed), then drain fully	No leak, no visible deformation; gauge calibrated; test pressure ≥ 1.3 × design 1.95 MPa g	DC3-V1-ME-EQL-001	☐	
HT-003	S10	V-201 shell	Hydrotest at 1.755 MPa g (top), hold 30 min (assumed), then drain fully	No leak, no visible deformation; gauge calibrated; test pressure ≥ 1.3 × design 1.35 MPa g	DC3-V1-ME-EQL-001	☐	
HT-004	S30	V-202 shell	Hydrotest at 2.470 MPa g (top), hold 30 min (assumed), then drain fully	No leak, no visible deformation; gauge calibrated; test pressure ≥ 1.3 × design 1.90 MPa g	DC3-V1-ME-EQL-001	☐	
HT-005	S40	E-201 shell (LP steam)	Hydrotest at 0.910 MPa g (top), hold 30 min (assumed), then drain fully	No leak, no visible deformation; gauge calibrated; test pressure ≥ 1.3 × design 0.70 MPa g	DC3-V1-ME-EQL-001	☐	
HT-006	S10	E-201 tubes (feed)	Hydrotest at 3.705 MPa g (top), hold 30 min (assumed), then drain fully	No leak, no visible deformation; gauge calibrated; test pressure ≥ 1.3 × design 2.85 MPa g	DC3-V1-ME-EQL-001	☐	
HT-007	S30	E-202 shell (overhead)	Hydrotest at 2.470 MPa g (top), hold 30 min (assumed), then drain fully	No leak, no visible deformation; gauge calibrated; test pressure ≥ 1.3 × design 1.90 MPa g	DC3-V1-ME-EQL-001	☐	
HT-008	S50	E-202 tubes (cooling water)	Hydrotest at 0.910 MPa g (top), hold 30 min (assumed), then drain fully	No leak, no visible deformation; gauge calibrated; test pressure ≥ 1.3 × design 0.70 MPa g	DC3-V1-ME-EQL-001	☐	

Item	Sys	Equipment / line / loop	Activity	Acceptance	Ref.	Done	Date / sign
HT-009	S40	E-203 shell (LP steam)	Hydrotest at 0.910 MPa g (top), hold 30 min (assumed), then drain fully	No leak, no visible deformation; gauge calibrated; test pressure ≥ 1.3 × design 0.70 MPa g	DC3-V1-ME-EQL-001	☐	
HT-010	S20	E-203 tubes (C-201 bottoms)	Hydrotest at 2.535 MPa g (top), hold 30 min (assumed), then drain fully	No leak, no visible deformation; gauge calibrated; test pressure ≥ 1.3 × design 1.95 MPa g	DC3-V1-ME-EQL-001	☐	
HT-011	S10	P-201A/B casing	Hydrotest at 3.705 MPa g (top), hold 30 min (assumed), then drain fully	No leak, no visible deformation; gauge calibrated; test pressure ≥ 1.3 × design 2.85 MPa g	DC3-V1-ME-EQL-001	☐	
HT-012	S30	P-202A/B casing	Hydrotest at 3.380 MPa g (top), hold 30 min (assumed), then drain fully	No leak, no visible deformation; gauge calibrated; test pressure ≥ 1.3 × design 2.60 MPa g	DC3-V1-ME-EQL-001	☐	
HT-013	S10	TP-10-01 (6"-P-200, 6"-P-201)	Hydrotest at 2.025 MPa g (B31.3: 1.5 × 1.35), 10 min hold, then drain	No leak at joints; test pack signed; blinds and temporary supports removed	DC3-V1-PE-LIN-001	☐	
HT-014	S10	TP-10-02 (4"-P-202, 6"-P-203)	Hydrotest at 4.275 MPa g (B31.3: 1.5 × 2.85), 10 min hold, then drain	No leak at joints; test pack signed; blinds and temporary supports removed	DC3-V1-PE-LIN-001	☐	
HT-015	S20	TP-20-01 (8"-P-211, 10"-P-212)	Hydrotest at 2.925 MPa g (B31.3: 1.5 × 1.95), 10 min hold, then drain	No leak at joints; test pack signed; blinds and temporary supports removed	DC3-V1-PE-LIN-001	☐	
HT-016	S30	TP-30-01 (8"-P-204, 6"-P-205, 1"-P-206, 6"-P-207)	Hydrotest at 2.850 MPa g (B31.3: 1.5 × 1.90), 10 min hold, then drain	No leak at joints; test pack signed; blinds and temporary supports removed	DC3-V1-PE-LIN-001	☐	
HT-017	S30	TP-30-02 (4"-P-208, 4"-P-209, 3"-P-213)	Hydrotest at 3.900 MPa g (B31.3: 1.5 × 2.60), 10 min hold, then drain	No leak at joints; test pack signed; blinds and temporary supports removed	DC3-V1-PE-LIN-001	☐	
HT-018	S30	TP-30-03 (4"-P-210)	Hydrotest at 2.925 MPa g (B31.3: 1.5 × 1.95), 10 min hold, then drain	No leak at joints; test pack signed; blinds and temporary supports removed	DC3-V1-PE-LIN-001	☐	
HT-019	S40	TP-40-01 (6"-LS-301, 6"-LS-302)	Hydrotest at 1.050 MPa g (B31.3: 1.5 × 0.70), 10 min hold, then drain	No leak at joints; test pack signed; blinds and temporary supports removed	DC3-V1-PE-LIN-001	☐	
HT-020	S40	TP-40-02 (2"-CD-501, 2"-CD-502)	Hydrotest at 1.050 MPa g (B31.3: 1.5 × 0.70), 10 min hold, then drain	No leak at joints; test pack signed; blinds and temporary supports removed	DC3-V1-PE-LIN-001	☐	
HT-021	S50	TP-50-01 (10"-CW-401, 10"-CW-402)	Hydrotest at 1.050 MPa g (B31.3: 1.5 × 0.70), 10 min hold, then drain	No leak at joints; test pack signed; blinds and temporary supports removed	DC3-V1-PE-LIN-001	☐	

DR · Drying (3 items)

Item	Sys	Equipment / line / loop	Activity	Acceptance	Ref.	Done	Date / sign
DR-001	S10	Feed system	Drain, then blow with dry air or N_2 until the outlet dew point is reached	Outlet dew point ≤ −20 °C (assumed) at every low point	DC3-V1-PID-001	☐	
DR-002	S20	Column and reboiler	Drain, then blow with dry air or N_2 until the outlet dew point is reached	Outlet dew point ≤ −20 °C (assumed) at every low point	DC3-V1-PID-001	☐	
DR-003	S30	Overhead and reflux	Drain, then blow with dry air or N_2 until the outlet dew point is reached	Outlet dew point ≤ −20 °C (assumed) at every low point	DC3-V1-PID-001	☐	

PG · N_2 purge (O_2 < 2 %) (3 items)

Item	Sys	Equipment / line / loop	Activity	Acceptance	Ref.	Done	Date / sign
PG-001	S10	Feed system	N_2 pressure-swing purge (calculator in part 6)	O_2 < 2 mol % at every high point and dead leg (portable analyzer)	SOP-C201-01	☐	
PG-002	S20	Column and reboiler	N_2 pressure-swing purge (calculator in part 6)	O_2 < 2 mol % at every high point and dead leg (portable analyzer)	SOP-C201-01	☐	
PG-003	S30	Overhead and reflux	N_2 pressure-swing purge (calculator in part 6)	O_2 < 2 mol % at every high point and dead leg (portable analyzer)	SOP-C201-01	☐	

LT · Tightness test (5 items)

Item	Sys	Equipment / line / loop	Activity	Acceptance	Ref.	Done	Date / sign
LT-001	S10	Feed system	N_2 tightness test at 0.80 MPa g (limited by the N_2 header), snoop every flange	Pressure decay ≤ 1 % per hour (assumed); no bubbles	DC3-V1-PID-001	☐	
LT-002	S20	Column and reboiler	N_2 tightness test at 0.80 MPa g (limited by the N_2 header), snoop every flange	Pressure decay ≤ 1 % per hour (assumed); no bubbles	DC3-V1-PID-001	☐	

Item	Sys	Equipment / line / loop	Activity	Acceptance	Ref.	Done	Date / sign
LT-003	S30	Overhead and reflux	N_2 tightness test at 0.80 MPa g (limited by the N_2 header), snoop every flange	Pressure decay ≤ 1 % per hour (assumed); no bubbles	DC3-V1-PID-001	☐	
LT-004	S40	LP steam and condensate	Service test with steam at the header pressure 0.45 MPa g	No leak at flanges and traps	DC3-V1-PE-LIN-001	☐	
LT-005	S50	Cooling water	Service test with CW at the supply pressure 0.45 MPa g	No leak; FI-216 reads the expected flow	DC3-V1-PE-LIN-001	☐	

VS · Valve stroke and fail position (10 items)

Item	Sys	Equipment / line / loop	Activity	Acceptance	Ref.	Done	Date / sign
VS-001	S70	FV-201	Stroke 0–25–50–75–100 % from the console; remove the air (or de-energise) and watch the travel	Fail closed as the INL; full travel; position feedback correct	DC3-V1-IN-INL-001	☐	
VS-002	S70	TV-202	Stroke 0–25–50–75–100 % from the console; remove the air (or de-energise) and watch the travel	Fail closed as the INL; full travel; position feedback correct	DC3-V1-IN-INL-001	☐	
VS-003	S70	FV-204	Stroke 0–25–50–75–100 % from the console; remove the air (or de-energise) and watch the travel	Fail open as the INL; full travel; position feedback correct	DC3-V1-IN-INL-001	☐	
VS-004	S70	TV-208	Stroke 0–25–50–75–100 % from the console; remove the air (or de-energise) and watch the travel	Fail closed as the INL; full travel; position feedback correct	DC3-V1-IN-INL-001	☐	
VS-005	S70	LV-202	Stroke 0–25–50–75–100 % from the console; remove the air (or de-energise) and watch the travel	Fail closed as the INL; full travel; position feedback correct	DC3-V1-IN-INL-001	☐	
VS-006	S70	LV-203	Stroke 0–25–50–75–100 % from the console; remove the air (or de-energise) and watch the travel	Fail closed as the INL; full travel; position feedback correct	DC3-V1-IN-INL-001	☐	
VS-007	S70	PV-210	Stroke 0–25–50–75–100 % from the console; remove the air (or de-energise) and watch the travel	Fail open as the INL; full travel; position feedback correct	DC3-V1-IN-INL-001	☐	
VS-008	S70	XV-201	Stroke 0–25–50–75–100 % from the console; remove the air (or de-energise) and watch the travel	Fail closed as the INL; full travel; position feedback correct; stroke ≤ 2 s (SRS)	DC3-V1-IN-INL-001	☐	
VS-009	S70	XV-202	Stroke 0–25–50–75–100 % from the console; remove the air (or de-energise) and watch the travel	Fail closed as the INL; full travel; position feedback correct; stroke ≤ 2 s (SRS)	DC3-V1-IN-INL-001	☐	
VS-010	S70	XV-209	Stroke 0–25–50–75–100 % from the console; remove the air (or de-energise) and watch the travel	Fail closed as the INL; full travel; position feedback correct; stroke ≤ 2 s (SRS)	DC3-V1-IN-INL-001	☐	

LC · Instrument loop check (42 items)

Item	Sys	Equipment / line / loop	Activity	Acceptance	Ref.	Done	Date / sign
LC-001	S70	FIC-201	Loop check from the field device to the console: FT-201, FV-201 (AI/AO)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-002	S70	TIC-202	Loop check from the field device to the console: TT-202, TV-202 (AI/AO)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-003	S70	FIC-204	Loop check from the field device to the console: FT-204, FV-204 (AI/AO)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-004	S70	TIC-227	Loop check from the field device to the console: TT-227 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-005	S70	TIC-208	Loop check from the field device to the console: TT-208, TV-208 (AI/AO)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-006	S70	LIC-202	Loop check from the field device to the console: LT-202, LV-202 (AI/AO)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-007	S70	LIC-203	Loop check from the field device to the console: LT-203, LV-203 (AI/AO)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-008	S70	PIC-210	Loop check from the field device to the console: PT-210, PV-210 (AI/AO)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-009	S70	LIC-201	Loop check from the field device to the console: LT-201 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	

Item	Sys	Equipment / line / loop	Activity	Acceptance	Ref.	Done	Date / sign
LC-010	S70	TI-201	Loop check from the field device to the console: TT-201 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-011	S70	FI-206	Loop check from the field device to the console: FT-206 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-012	S70	PI-207	Loop check from the field device to the console: PT-207 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-013	S70	TI-241	Loop check from the field device to the console: TT-241 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-014	S70	TI-242	Loop check from the field device to the console: TT-242 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-015	S70	TI-243	Loop check from the field device to the console: TT-243 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-016	S70	TI-245	Loop check from the field device to the console: TT-245 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-017	S70	TI-246	Loop check from the field device to the console: TT-246 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-018	S70	TI-209	Loop check from the field device to the console: TT-209 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-019	S70	PI-211	Loop check from the field device to the console: PT-211 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-020	S70	PDI-212	Loop check from the field device to the console: PDT-212 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-021	S70	FI-205	Loop check from the field device to the console: FT-205 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-022	S70	PI-205	Loop check from the field device to the console: PT-205 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-023	S70	FI-202	Loop check from the field device to the console: FT-202 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-024	S70	AI-220	Loop check from the field device to the console: AT-220 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-025	S70	FI-203	Loop check from the field device to the console: FT-203 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-026	S70	AI-219	Loop check from the field device to the console: AT-219 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-027	S70	TI-213	Loop check from the field device to the console: TT-213 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-028	S70	FI-216	Loop check from the field device to the console: FT-216 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-029	S70	TI-214	Loop check from the field device to the console: TT-214 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-030	S70	TI-215	Loop check from the field device to the console: TT-215 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-031	S70	FI-217	Loop check from the field device to the console: FT-217 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-032	S70	LSLL-202	Loop check from the field device to the console: LS-202 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-033	S70	LSHH-203	Loop check from the field device to the console: LS-203 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-034	S70	ESD	Loop check from the field device to the console: HS-ESD, HS-RST (DI/DI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	

Item	Sys	Equipment / line / loop	Activity	Acceptance	Ref.	Done	Date / sign
LC-035	S70	PDSL-230	Loop check from the field device to the console: PDT-230 (AI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-036	S70	XV-201	Loop check from the field device to the console: XV-201, ZSC-201 (DO/DI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-037	S70	XV-202	Loop check from the field device to the console: XV-202, ZSC-202 (DO/DI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-038	S70	XV-209	Loop check from the field device to the console: XV-209, ZSC-209 (DO/DI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-039	S70	P-201A	Loop check from the field device to the console: HY-201A, YI-201A, XA-201A (DO/DI/DI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-040	S70	P-201B	Loop check from the field device to the console: HY-201B, YI-201B (DO/DI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-041	S70	P-202A	Loop check from the field device to the console: HY-202A, YI-202A, XA-202A (DO/DI/DI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	
LC-042	S70	P-202B	Loop check from the field device to the console: HY-202B, YI-202B, XA-202B (DO/DI/DI)	Signal at 0, 50, 100 % of range (or on/off) on the console; tag, range and alarms as the INL	DC3-V1-IN-INL-001	☐	

PS · PSV bench test and set certificate (1 items)

Item	Sys	Equipment / line / loop	Activity	Acceptance	Ref.	Done	Date / sign
PS-001	S60	PSV-201	Bench test of the set pressure (cold differential test pressure 1.80 MPa g); seat tightness at 90 % of set; tag and seal	Pops at 1.746–1.854 MPa g (± 3 %); API 527 leak rate; set certificate filed	DC3-V1-PE-RLB-001	☐	

MB · Motor bump test / turbine solo run (4 items)

Item	Sys	Equipment / line / loop	Activity	Acceptance	Ref.	Done	Date / sign
MB-001	S80	P-201A motor 75 kW	Insulation test, bump for rotation (uncoupled), 2 h solo run (assumed)	Direction of rotation correct; no-load current and vibration normal; insulation ≥ 100 MΩ (assumed)	DC3-V1-EE-LL-001	☐	
MB-002	S80	P-202A motor 30 kW	Insulation test, bump for rotation (uncoupled), 2 h solo run (assumed)	Direction of rotation correct; no-load current and vibration normal; insulation ≥ 100 MΩ (assumed)	DC3-V1-EE-LL-001	☐	
MB-003	S80	P-202B motor 30 kW	Insulation test, bump for rotation (uncoupled), 2 h solo run (assumed)	Direction of rotation correct; no-load current and vibration normal; insulation ≥ 100 MΩ (assumed)	DC3-V1-EE-LL-001	☐	
MB-004	S40	P-201B steam turbine	Uncoupled solo run on LP steam, overspeed trip test, governor check; auto-start from HS-201B	Trips at 110 % of rated speed (assumed); starts from the DCS / FSLL-201	DC3-V1-ME-DS-001	☐	

SF · SIS function test (6 items)

Item	Sys	Equipment / line / loop	Activity	Acceptance	Ref.	Done	Date / sign
SF-001	S70	HS-ESD — Emergency shutdown push-button (DCS / SIS screen)	Function test with the real sensor (or a calibrated input at the switch) and the real final elements	Effects as the C&E: XV-201 C, FIC-201 MAN, XV-202 C, XV-209 C, TIC-202 MAN, TIC-208 MAN, SIS TP, 1ST F, RED A, AUD A; delay 0 s	DC3-V1-IN-CE-001	☐	
SF-002	S70	FSLL-201 — Feed flow low-low (FT-201): spare feed pump P-201B auto-start	Function test with the real sensor (or a calibrated input at the switch) and the real final elements	Effects as the C&E: P-201B ST, AMB A, AUD A; delay 5 s	DC3-V1-IN-CE-001	☐	
SF-003	S70	LSLL-202 — C-201 sump level low-low trip (LS-202): heat off	Function test with the real sensor (or a calibrated input at the switch) and the real final elements	Effects as the C&E: XV-202 C, XV-209 C, TIC-202 MAN, TIC-208 MAN, SIS TP, 1ST F, RED A, AUD A; delay 5 s	DC3-V1-IN-CE-001	☐	
SF-004	S70	LSHH-203 — Reflux drum V-202 level high-high trip (LS-203): feed and heat off	Function test with the real sensor (or a calibrated input at the switch) and the real final elements	Effects as the C&E: XV-201 C, FIC-201 MAN, XV-202 C, XV-209 C, TIC-202 MAN, TIC-208 MAN, SIS TP, 1ST F, RED A, AUD A; delay 5 s	DC3-V1-IN-CE-001	☐	
SF-005	S70	PDSL-230 — Reflux pump differential pressure low (PDT-230): spare reflux pump P-202B auto-start	Function test with the real sensor (or a calibrated input at the switch) and the real final elements	Effects as the C&E: P-202B ST, AMB A, AUD A; delay 3 s	DC3-V1-IN-CE-001	☐	

Item	Sys	Equipment / line / loop	Activity	Acceptance	Ref.	Done	Date / sign
SF-006	S70	I-201 — Open XV-209 / XV-202 refused: start permissive not met	Function test with the real sensor (or a calibrated input at the switch) and the real final elements	Effects as the C&E: SIS RS, AMB A	DC3-V1-IN-CE-001	☐	

6 · Commissioning and start-up mapped to the OTS procedure

Steps of the simulator's cold start-up procedure (practised by the operators before the first feed) against the commissioning activities and the SOP tab.

#	Commissioning / start-up activity	OTS start-up step	SOP	Where in the OTS
1	Pre-startup safety review (PSSR) signed; pre-commissioning dossier and punch A closed for S10–S80	before the OTS cold state	SOP-C201-01 (pre-requisites)	not in the OTS (field and paperwork)
2	Utilities commissioned: cooling water, LP steam, instrument air, N_2, fuel-gas header; the SIS tested (part 7, SF rows)	before the OTS cold state	SOP-C201-01 steps 5, 8	not in the OTS (the OTS starts with utilities available)
3	Gassing-up: N_2 displaced by fuel gas, the unit left under a fuel-gas blanket (cold state of the OTS)	before the OTS cold state	SOP-C201-01 step 3	the OTS cold state
4	Heat sink before heat: CW through E-202 and the vent path to fuel gas open	1 Open the cooling-water block HV-216 and the vent block HV-210	SOP-C201-02 step 1	Field / P&ID: HV-216, HV-210
5	First LPG into the column: feed pump started, column filled to the reboiler-cover level	2 Fill the column: FIC-201 in MAN and a feed pump running 3 Sump level LIC-202 at 40 % or more	SOP-C201-02 steps 2–3	P&ID: P-201A, FIC-201, LIC-202
6	Feed rate agreed with the upstream unit (inventory balance of V-201)	5 Upstream supplying V-201 at the rate you feed (within 5 m^3/h)	SOP-C201-02 step 3	P&ID: FIC-201, LI-201 / LG-201
7	Heat on through the SIS start permissives	4 Enable heating on the SIS with TIC-208 in MAN	SOP-C201-02 steps 4–5	SIS display
8	Start-up gas vented while the pressure builds; then pressure control in AUTO	7 Vent the start-up gas while the pressure builds (PV-210 in MAN) 8 PIC-210 in AUTO at 1.70 MPa g	SOP-C201-02 steps 6, 8	P&ID: PV-210, PIC-210
9	Total reflux established: reflux pump on, FIC-204 in AUTO	6 Reflux established: a reflux pump running and FIC-204 in AUTO	SOP-C201-02 step 7	P&ID: P-202A, FIC-204
10	Bottoms to the debutanizer on level control	9 Bottoms on LIC-202 in AUTO	SOP-C201-02 step 9	P&ID: LIC-202
11	Stripping section on temperature control	10 TIC-208 in AUTO with its set point at 76.9 ±1 °C	SOP-C201-02 step 10	P&ID: TIC-208
12	Preheat and propane draw in AUTO: first product to storage	11 TIC-202 and LIC-203 in AUTO	SOP-C201-02 step 11	P&ID: TIC-202, LIC-203
13	Ramp to the design rate with the upstream unit matched	12 Feed at 60 ±2 m^3/h in AUTO with the upstream matched	SOP-C201-02 step 12	Trends: FI-201, PDI-212
14	Line-out on spec: ready for the performance test	13 Hold the line-out for 10 minutes: both products on spec, pressure 1.70 ±0.03 MPa g	SOP-C201-02 step 13	Part 9 of this tab (live guarantees)

7 · Performance test against the HMB design case

72 h at the guarantee capacity with both products on spec, instruments calibrated beforehand, corrections for cooling-water temperature and feed composition agreed in the contract (query EC-08). The design case of the simulator meets every guarantee; the page checks the running plant live.

Guarantee	Required	Design case (model)	Result
Capacity: feed FI-201	≥ 59.4 m^3/h (design 60 m^3/h = 31.8 t/h)	60.0 m^3/h	✓
Propane product C4+, AI-219	≤ 2.5 mol %	1.50 mol %	✓
Bottoms C3, AI-220	≤ 2.0 mol %	1.00 mol %	✓
LP steam per tonne of feed (FI-205 + FI-206)	≤ 0.317 t/t (design 0.302 + 5 %)	0.302 t/t	✓
Cooling water FI-216	≤ 346 m^3/h	330 m^3/h	✓
Test condition: PIC-210, no relief	1.70 ± 0.03 MPa g; PSV-201 closed	1.700 MPa g	✓

8 · Assumptions

- Equipment hydrotest ASME VIII Div. 1 UG-99(b): 1.3 × design pressure × stress ratio, with MAWP taken = design pressure and the stress ratio 1 (assumed); piping ASME B31.3 §345.4.2: 1.5 × design × 1. Hold 30 min (vessels) / 10 min (piping), assumed.
- C-201 vertical field test: water head over T/T + D/2 = 23.82 m; bottom hoop stress limited to 0.9 × S_y = 0.9 × 260 MPa (SA-516 Gr. 70, assumed limit).
- Dryness: outlet dew point ≤ −20 °C (assumed, query EC-06). Oxygen < 2 mol % before hydrocarbon; N_2 supply 0.1 mol % O_2 at 30 °C; system volumes = vessel volumes + 15 % for exchangers and piping (assumed).
- N_2 header 0.80 MPa g (assumed): tightness test at the lower of the N_2 header, the maximum operating pressure and 90 % of the PSV-201 set; pressure decay ≤ 1 % per hour.
- PSV-201 cold differential test pressure = set 1.80 MPa g (no superimposed back-pressure, temperature correction neglected); set tolerance ± 3 %; seat tightness API 527 at 90 % of set.
- Motor insulation ≥ 100 MΩ, 2 h solo run; P-201B turbine overspeed trip at 110 % of rated speed (assumed). Water run-in from a temporary tank with 3 m static suction (assumed).
- Performance guarantees: capacity −1 % (meter tolerance), LP steam per tonne of feed +5 %, cooling water +5 % on the HMB design case; specification limits AI-219 2.5 mol %, AI-220 2.0 mol % (assumed tolerances, query EC-08).

Engineering queries raised

No.	Subject	Query	Proposed answer	Status
EC-01	P-201A/B run-in on water	Run on water, P-201 develops the same head (m) but 1.82 × the pressure and power of LPG (ρ 548.5 kg/m³): at 60 m³/h it absorbs 90.7 kW against a 75 kW motor, and its shut-off pressure on water is 4.22 MPa g against the 2.85 MPa g design of the casing, E-201 tubes and the 4"-P-202 / 6"-P-203 lines (hydrotest 3.705 MPa g).	No water run-in against the plant system. Run P-201A in on water only on a temporary loop with a full-flow relief set below 2.85 MPa g and the flow limited to 46 m³/h (motor), or commission the pumps directly on LPG after the solo run of the motor.	OPEN
EC-02	P-202A/B run-in on water	At 81 m³/h on water P-202 would absorb 43.9 kW (2.12 × the 20.7 kW on reflux at ρ 471.7 kg/m³) against a 30 kW motor; shut-off on water 1.51 MPa g is inside the 2.60 MPa g design.	Limit any water run to 47 m³/h by throttling the discharge, with the motor current watched; preferably commission on propane.	OPEN
EC-03	C-201 hydrotest weight for the foundation and skirt	Full of water C-201 weighs 76.6 t (empty 23.2 t + 53.4 m³ water) against about 27.6 t in operation (model holdup 4.4 t). A vertical field hydrotest also raises the bottom to 2.704 MPa g (water head 0.234 MPa over 23.82 m).	Design the foundation, skirt and anchor bolts for the hydrotest load case; test C-201 in the shop horizontally at 2.535 MPa g and in the field only after a repair, with civil confirmation.	OPEN
EC-04	Tightness test pressure	The N_2 tightness test is limited to 0.80 MPa g by the N_2 header (assumed); a test at the operating pressure (up to 1.740 MPa g in the column system) would also exceed 90 % of the PSV-201 set (1.62 MPa g) because the set is only 5.9 % above operating (EP-01).	N_2 test at the header pressure; the range above it is proven by a hydrocarbon leak check at each pressure step of the start-up (hot bolting rounds). Revisit when EP-01 (PSV set) is answered.	OPEN
EC-05	Long-lead items on the critical path	With the assumed lead times PKG-01 Column C-201 (44 weeks, float 2 weeks) and PKG-04 Pumps P-201A/B, P-202A/B (48 weeks, float 2 weeks) have ≤ 4 weeks of float to their construction need dates; one month of delay in their purchase orders delays mechanical completion.	Place the column and pump purchase orders on the FEED datasheets (DC3-V1-ME-DS-001) with a later technical freeze, expedite monthly, and plan the column erection as the first heavy lift.	OPEN
EC-06	Dryness criterion after the hydrotest	The design basis has no dryness criterion; −20 °C dew point is assumed. The feed arrives at −1 °C and propane auto-refrigerates to −42 °C on depressuring: free water would form ice / hydrates in valves and instrument taps.	Confirm the dew point (−20 °C or lower) with the propane product water specification; dry with dry N_2 after the hydrotest.	OPEN
EC-07	PSHH-221 in the SIS FAT scope	PSHH-221 is a recommended SIF (HAZOP, LOPA, SRS) but not in the logic; the SIS FAT tests 6 causes. Adding it after the FAT means a second FAT and re-validation.	Decide on PSHH-221 (PT-221, own nozzle) before the SIS configuration freeze so it is tested at the FAT.	OPEN
EC-08	Performance-test corrections and tolerances	The guarantees are stated at the design case (cooling water 330 m³/h at 15 °C, LP steam 0.45 MPa g, feed z = 0.5707); on the test day these differ. Tolerances assumed: capacity −1 %, steam +5 %.	Agree in the contract the correction curves (CW temperature, feed composition) and the measurement uncertainty; calibrate FI-201, FI-205, FI-206 and the analyzers before the test.	OPEN
EC-09	OTS start-up begins under fuel gas	The simulator's start-up procedure starts from the cold state under a fuel-gas blanket: N_2 purging, gassing-up and the oxygen check of SOP-C201-01 are not in the simulator.	Keep them as field steps in the commissioning procedure; brief the operators on the OTS for the steps after gassing-up only.	OPEN